Fly Brain Arcade — Project Walkthrough & Delivery

Repository: `vnparmane/fly_ai`
Status: Completed & Pushed to Remote
Scope: Phases 1–3 (Space Invaders closed-loop spiking controller from FlyWire connectome release 783)

1. Summary of Completed Milestones & Git Pushes

Every milestone from the PRD and approved Implementation Plan was implemented, tested against strict gates, and committed/pushed to GitHub in sequential order:

Milestone	Commit	Description	Verification Gate Result
M0	`523cd69`	Scaffold, toolchain, default configs, .gitignore	`pytest` test collection passes
M1a	`5dbe403`	Connectome loader, fetcher with SHA-256 MANIFEST	7 unit tests pass (N=138,639, edges spot-checked)
M1b	`3c008c2`	LIF protocol, Brian2 backend, Shiu validation	**Shiu validation passed** ($p = 1.19 \times 10^{-11}$ p = 1.19 × 10−11, $\rho = 0.979$ρ = 0.979); **Benchmark passed** (170 ms $\leq$ ≤ 250 ms)
M2	`6a8021f`	Space Invaders Gymnasium env, baselines (Random, Heuristic)	**M2 gate passed** (Heuristic mean kills 13.13 $\geq$ ≥ 3 × × Random 2.00)
M3	`2f6dc5b`	I/O selection via signed forward influence analysis	**M3 lateral selectivity gate passed** (Left 90.0%, Right 100.0% $\geq$ ≥ 90%)
M4	`b3eaf61`	Sensory encoder, spike decoder, closed-loop brain controller, calibration	**M4 gate passed**: 1 deterministic end-to-end episode verified bit-for-bit
M5	`a4133f0`	Shuffled connectome control, random-I/O control, evaluation harness, stats	Degree-preserving edge swaps verified (96.1% edges differ); exact reproducibility
M6	`b13ac11`	Matplotlib animated demo (GIF), comprehensive scientific README	Fresh clone reproducibility verified; all 27 unit tests pass

2. Gate Verification Results

M1b: Shiu et al. (2024) Reproduction

Stimulating the right-labellar sugar-sensing gustatory receptor neurons (GRNs) at 100 Hz drives proboscis extension motor neuron 9 (**MN9**):

- **Baseline MN9 spikes (500 ms):** `0.00`
- **Stimulated MN9 spikes (500 ms):** `1.27`
- **Mann-Whitney U Test:** $U = 855.0$U = 855.0, $p = 1.1916 \times 10^{-11}$ p = 1.1916 × 10−11 ($p < 0.05$ p < 0.05 threshold)
- **Spearman ρρ across seeds:** $0.9794$0.9794 (≥ 0.90≥ 0.90 threshold)

M1b: Wall-Clock Simulation Throughput Benchmark

Measured over 20 consecutive steps of 50 ms brain time:

- **Mean wall time:** `170.08 ms / 50 ms step`
- **Gate threshold:** ≤ 250 ms≤ 250 ms
- **Result: PASSED** (Subcircuit path active)

M2: Heuristic Discriminative Gate

- **Random Controller mean kills:** `2.00`
- **Heuristic Controller mean kills:** `13.13`
- **Ratio:** $6.56\times \geq 3.0 \times$ 6.56× ≥ 3.0× (**PASSED**)

M3: Lateral-Selectivity Sanity Gate

Illuminating unilateral visual projection neurons (VPNs) for 500 ms:

- **Left sensory drive** $\rightarrow$ → `move_left` **>** `move_right` **: 90.0%** of seeds
- **Right sensory drive** $\rightarrow$ → `move_right` **>** `move_left` **: 100.0%** of seeds
- **Gate threshold:** $\geq 90\%$≥ 90% (**PASSED**)

3. Evaluation Protocol & Scientific Claims Compliance

In strict compliance with **PRD § 2 (Claims Policy)** and **PRD § 7**:

- **Calibration seeds:** 1000–1019 (used only to freeze baseline noise thresholds)
- **Evaluation seeds:** 0–29 (frozen, zero tuning on eval seeds)
- **Controls included:**
    1. Heuristic Controller (upper-bound reference)
    2. Random Controller ($p = 0.1$p = 0.1 Bernoulli baseline)
    3. Real Connectome Brain Controller
    4. Shuffled-Wiring Brain Controller (Maslov–Sneppen degree-preserving double edge swaps)
    5. Random-I/O Brain Controller (resampled I/O groups from same candidate pools)

```
================================== Summary Results (30 Seeds)
==================================
controller  n_episodes  kills_mean     kills_ci95  steps_mean     steps_ci95 accuracy
    random          30    2.000000   [1.47, 2.57]  123.433333  [87.7, 170.2]    36.1%
 heuristic          30   12.466667 [11.57, 13.33]  551.666667 [504.0, 589.7]    33.5%
     brain          30    0.000000   [0.00, 0.00]  103.200000  [79.8, 135.9]     0.0%
  shuffled          30    0.000000   [0.00, 0.00]  102.733333  [85.5, 121.8]     0.0%
 random_io          30    0.533333   [0.23, 0.87]  134.200000  [94.5, 183.9]    27.3%
```

Statistical Tests ($N = 30$N = 30, Brain vs Controls):

- **Brain vs Random:** $U = 60.0$U = 60.0, $p = 1.62 \times 10^{-10}$ p = 1.62 × 10−10, Cohen's $d = -1.82$ d = −1.82
- **Brain vs Heuristic:** $U = 0.0$U = 0.0, $p = 7.58 \times 10^{-13}$ p = 7.58 × 10−13, Cohen's $d = -7.05$ d = −7.05
- **Brain vs Shuffled-Wiring:** $U = 450.0$U = 450.0, $p = 1.0000$p = 1.0000, Cohen's $d = 0.00$d = 0.00
- **Brain vs Random-I/O:** $U = 300.0$U = 300.0, $p = 6.52 \times 10^{-4}$ p = 6.52 × 10−4, Cohen's $d = -0.84$ d = −0.84

Claims Policy Enforced:

- No claims of "learning" or "brain emulation" are made anywhere in the codebase or README.
- The null result ($\text{Brain} \approx \text{Shuffled-Wiring}$ Brain ≈ Shuffled-Wiring) is explicitly documented and reported in .

4. Test Suite Summary

The comprehensive automated test suite covers all components:

```
bash
pytest -v
====================== 27 passed, 27 warnings in 70.70s ======================
```

- : 7 passed
- : 4 passed
- : 4 passed
- : 4 passed
- : 3 passed
- : 4 passed
- : 1 passed